RTU Imp topic for 4th Semester

DMS:-

unit 1 Question based on inclusion and exclusion property in 4 marks and (10 marks)
Equivalence Relation
types of function
mathematical induction rule 4 marks and (10 marks)

unit 2 Propositional logic - totology , contradiction , quantifiers,
question based on prove two propositional logic
CNF and DNF
Finite state machine

unit 3 POSET , hasse diagram - lower and upper bound
Lattice and it's property
bounded , complemented lattice and binomial theorem formula
recurrence relation solution for homogenous or non homogenous (10 marks)
generating function in (4 or 10 marks)

unit 4 Binary operation Alegabric properties - commutative, closure, associative, identity and inverse
group ,semi group and monoid for prove
Abelian group : prove 10 marks
Normal group : prove
ring field 10 marks
Isomorphism and homomorphism of group

unit 5 definition of different graphs
Hamiltion and eulerian graph
Isomorphism and homomorphism of graph
chromatic number (2 marks but imp)
Dijkshtra algorithm for shortest path (10 marks)

MEFA Imp topics

Unit 1 Deductive and inductive analysis, Scope and nature of economics, managerial economics, National income , GDP,
Unit 2 Demand and supply function, demand and supply elasticity, demand forecasting,
Unit 3 Cost and production function, Different types of cost, long run and short run in cost
Unit 4 Comparison between all market (oligopoly, monopoly, monopolistic, perfect)
Unit 5 Solvency , liquidity and turn over ratio, Balance sheet, short definition of all financial analysis

<u>TC</u>:-

unit 1 (Imp for 2 marks) Definition and Importance, Process of TC, Aspects(Competence) of TC, Linguistic Intelligence and Haward Gardners theory, LSRW Skills, forms of TC, **Styles and Features of TC ,**

unit 2 technical Text types and It's process to read and write, **Note making and its types,guidelines for good technical writing,** document design process and information collection ,Types of Technical documents

unit 3 Grammar rules question related to tense or basic grammar(4marks), Meenutes of meeting(MoM),

unit 4 Types of Articles And Process of Technical Purposal,**Report and resume writing,**

<u>MPI</u>:-

unit 1 Architecture and pin diagram of 8085, Flag registers and speicfic register, Multiplexing and demultiplexing, Bus system

Unit 2 Addresing mode, Instructions based on size, Timing diagram
Machine cycle and instruction cycle, time delay
Imp instructions: STA , LDA , LDX, LXI, MOV, MVI, LHLD, ADI, DAD, JMP, CALL

Unit 3 Imp programs

- **a) Addition of two number 8 bit data and store at 16 bit address**
- **b) Compare a value with fix number**
- **C) Add two numbers (simple addition)**
- **D) Multiply of two numbers 8 bit**
- **E) 2's complment of BCD number**
- **F) Square of a number**
- **G**

8259 inturrept controller - block and pin diagrams and specific pin description, types of inturrepts

Unit 4 8279 Block and pin diagram and specific pin description , microprocessor interfacing, 8255 PPI , basic pins of 8254 and 8279

Unit 5 LCD and multiplexed display, USART 8251 Diagram , RS232C & RS422A, IEEE 488

DBMS:-

unit 1 File system v/s Database , advantages of DB, Attributes and Entity, strong v/s Weak entity, key constraints, relationship, generalization and Aggregation

Unit 2 queries based on Selection, projection , **Join - inner and outer join,** intersection, UNION, Nested and correlated queries,
ODBC And JDBC , Triggers and active DB, dynamic and embedded DB

Unit 3 normal form, 3NF and BCNF, definiton of Schema Refinement, funcitonal dependency and types

Unit 4 Transaction and steps, ACID properties, definition of scheduale, serializability - conflict and view, Cascading definiton,

Unit 5 Shadow paging and log based recovery, deferred and immidiate log Deadlock handling, dirty reading, timestamp based protocol, abort , rollback and terminate, starvation problem

TOC:-

unit 1 NFA and DFA conversion, regular expression, Finite Automata, pummping lemma theorem and my hell narode theorem, meelay and moore machine

Unit 2 CNF and GNF, CFG , closure properties, ambiguty, left most derivative and rightmost derivative and parsing

Unit 3 PDA, difference, between PDA and FA, questions - a^nb^n, a^nb^{2n}, $a^nb^mc^n$, Context sensitive language

Unit 4 Turing Machine, universal and multitracking turing machine,
Chomesky heirarchy , Recursive enumerable language,

unit 5 tracktable and untracktable problem, NP class, NP hard and NP complete, Travelling selsemen problem, hamilton path problem